

STA323 Assignment2

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Q1

本题我的实现思路是为每行创建 key-value 对，如果origin = ATL，则key = 0；其他情况下key = random(1, 2, 3)。然后再使用 partitionBy(4) 将数据分到 4 个分区

程序运行结果如下：

```
Total partitions: 4

Partition 0:
  Total records: 91484
  ATL records: 91484
  Other records: 0
  Sample origins: ['ATL', 'ATL', 'ATL', 'ATL', 'ATL', 'ATL', 'ATL', 'ATL', 'ATL', 'ATL']...

Partition 1:
  Total records: 433552
  ATL records: 0
  Other records: 433552
  Sample origins: ['ABE', 'ABE', 'ABE', 'ABE', 'ABE', 'ABE', 'ABE', 'ABE', 'ABE', 'ABE']...

Partition 2:
  Total records: 432924
  ATL records: 0
  Other records: 432924
  Sample origins: ['ABE', 'ABE', 'ABE', 'ABE', 'ABE', 'ABE', 'ABE', 'ABE', 'ABE', 'ABE']...

Partition 3:
  Total records: 433618
  ATL records: 0
  Other records: 433618
  Sample origins: ['ABE', 'ABE', 'ABE', 'ABE', 'ABE', 'ABE', 'ABE', 'ABE', 'ABE', 'ABE']...
```

Q2

(1)

程序运行结果如下：

```
Instructors schema: ['_class', 'id', 'title', 'name', 'display_name', 'job_title', 'image_50x50', 'image_100x100', 'initials', 'url']
Courses schema: ['id', 'title', 'url', 'rating', 'num_reviews', 'num_published_lectures', 'created', 'last_update_date', 'duration', 'instructors_id', 'image']

Total joined records: 83094

--- Sample joined records (first 5) ---
+-----+-----+-----+-----+-----+-----+
|instructors_id|course_title|name|display_name|rating|num_reviews|
+-----+-----+-----+-----+-----+-----+
|9685726|The Complete Python Bootcamp From Zero to Hero in Python|Jose|Jose Portilla|4.5927815|452973|
|31334738|The Complete 2023 Web Development Bootcamp|Dr. Angela|Dr. Angela Yu|4.667258|263152|
|4466306|The Web Developer Bootcamp 2023|Colt|Colt Steele|4.6961474|254711|
|13952972|Angular - The Complete Guide (2023 Edition)|Maximilian|Maximilian Schwarzmüller|4.5926924|180257|
|31334738|100 Days of Code: The Complete Python Pro Bootcamp for 2023|Dr. Angela|Dr. Angela Yu|4.6952515|177568|
+-----+-----+-----+-----+-----+-----+

only showing top 5 rows
Output saved to /data/assignment2/Q2_1_output
```

(2)

本小问的主要完成思路是先筛选课程标题包含 "spark" 且创建时间 > 2018-01-01 的课程，然后与 instructors 表进行 INNER JOIN，再按 course_rating 降序排序，取第一条（评分最高）。运行结果如下：

```
=====
SQL Query executed.
--- Result: Top-Rated Spark Instructor ---
+-----+-----+-----+-----+-----+
|display_name|job_title|course_rating|course_title|course_created|
+-----+-----+-----+-----+-----+
|Lara F|SHARING MY KNOWLEDGE WITH ENTHUSIASTIC LEARNERS!|5.0|Learn how to create Instagram beauty filters with Spark AR|2020-07-31T14:22:06Z|
+-----+-----+-----+-----+-----+

--- All matching Spark courses (for reference) ---
```

(3)

程序运行结果如下：

```
SQL Query executed.
Total matching courses: 347
--- All Interview Courses (sorted) ---
+-----+-----+-----+-----+-----+-----+-----+
|course_id|course_title|course_rating|course_created|num_reviews|duration|instructors_id|
+-----+-----+-----+-----+-----+-----+-----+
|4886926|Interview Oriented Data Structure Arrays & Linked List C/C++|5.0|2022-09-17T17:57:14Z|27|16.5 total hours|57923434|
|4309400|CATIA V5 FOR JOBS INTERVIEW|5.0|2021-09-20T12:54:23Z|5|7.5 total hours|173220222|
|4829150|Réaliser des interviews au rendu professionnel (PARTIE 2)|4.9|2022-08-12T14:54:06Z|9|2 total hours|79438676|
|4722894|"The "BigTech" System Design Interview Bootcamp"|4.9|2022-06-07T14:53:40Z|22|5.5 total hours|35934622|
|4499476|Power BI Interview Questions and Answers|4.9|2022-01-17T11:08:03Z|32|5.5 total hours|183242794|
|4266596|C#/.NET - 50 Essential Interview Questions (Junior Level)|4.9|2021-08-27T13:46:55Z|309|6.5 total hours|35184570|
|4189444|SAP MM Mock Interview Videos Part 1|4.9|2021-07-17T04:48:21Z|65|2 total hours|68817574|
|4149490|Mastering critical SKILLS for Coding Interviews C++: Part 1|4.9|2021-06-27T16:11:33Z|227|18 total hours|2994446|
|1493228|JavaScript Interview Prep: Practice Problems|4.9|2018-01-03T20:33:09Z|1078|4 total hours|23967306|
|4851774|Win your Product Management job interview with Big Tech's PM|4.8|2022-08-26T10:43:53Z|67|6.5 total hours|110412372|
|4607826|Docker & Kubernetes Interview Readiness Course|4.8|2022-03-22T12:41:09Z|75|1.5 total hours|59772743|
|4562744|C#/.NET - 50 Essential Interview Questions (Mid Level)|4.8|2022-02-21T16:40:51Z|136|8 total hours|35184570|
|3872994|Java Multithreading & Concurrency - Interview Practice Exams|4.8|2021-02-24T18:28:10Z|38|101 questions|44831418|
|3588944|Coding Interview Bootcamp: Algorithms (Java, Python, C++)|4.8|2020-10-23T19:48:55Z|39|13 total hours|137843512|
|3366332|Salesforce Business Analyst Interviews Questions & Answers|4.8|2020-07-26T09:38:02Z|178|1 total hour|127617236|
|2537130|Dynamic Programming Python, Coding Interviews & Applications|4.8|2019-08-31T16:29:28Z|81|5 total hours|41149680|
|1927620|The Real Estate Financial Modeling Interview Exam Guide|4.8|2018-09-23T04:19:22Z|787|10 total hours|5020122|
|1386504|Get your JAVA dream job! Collection Interview Questions|4.8|2017-10-08T18:14:24Z|29|3.5 total hours|19485006|
|1157988|Get your Java dream job! Beginners interview preparation|4.8|2017-03-25T22:54:38Z|223|6.5 total hours|19485006|
|816152|Algorithms Data Structures in Java #2 (+INTERVIEW QUESTIONS)|4.8|2016-04-08T06:35:28Z|770|13 total hours|837576|
+-----+-----+-----+-----+-----+-----+-----+

only showing top 20 rows
```

Q3

两小问程序运行结果如下：

26/04/11 02:02:53 INFO DAGScheduler: Job 53 finished: collect at /data/assignment2/Q3_data/Q3_1.py:107, took 36.337482 ms

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RESULTS:

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SME average: 977068491.69
LAM average: 925527784.43
KAM average: 1090161479.60

SME - LAM difference: 51540707.26
LAM - KAM difference: -164633695.17
KAM - SME difference: 113092987.91

26/04/11 02:02:02 INFO DAGScheduler: Job 20 finished: collect at /data/assignment2/Q3_data/Q3_2.py:140, took 44.767085 ms

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RESULTS:

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SME average: 977068491.69
LAM average: 925527784.43
KAM average: 1090161479.60

SME - LAM difference: 51540707.26
LAM - KAM difference: -164633695.17
KAM - SME difference: 113092987.91

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